

ECE 215: Circuit Theory and Devices (CTD)

End-Sem Exam – Set B

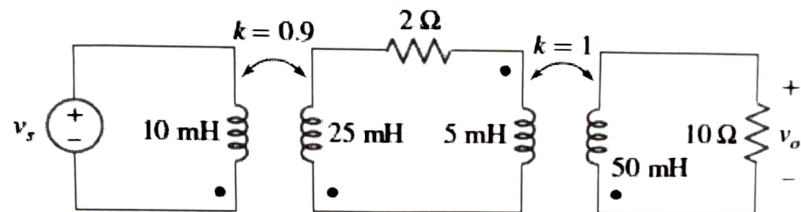
Date: December 02, 2025	Duration: 2 hours	Total points: 50 (+10 bonus)
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Instructions

- Write your set in the answer sheet and Bode plot sheet.
- Remember to write units in all your answers.
- Clearly mention the question number you are attempting. Otherwise, zero marks will be given.
- Sharing of calculators and stationery is strictly prohibited.

Question 1 [CO1, CO2] [5 points]

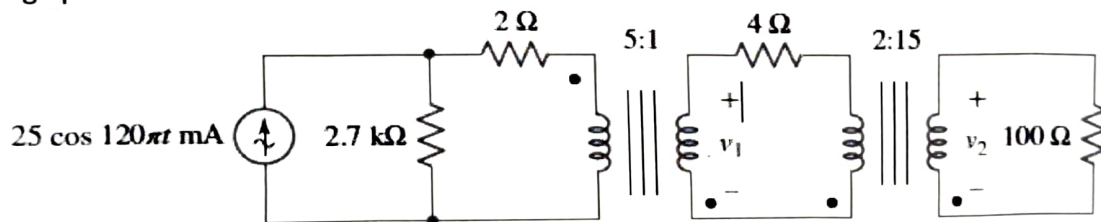
For the following circuit, write mesh current equations in terms of I_1 , I_2 and I_3 .



Questions 2 [CO1, CO2] [10 points]

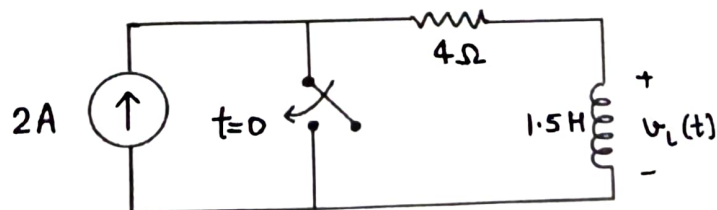
With respect to the circuit depicted below, calculate

- voltages v_1 and v_2
- average power delivered to each resistor.



Questions 4 [CO1, CO3] [10 points]

In the following circuit, the switch is turned on at $t = 0$. Find the voltage across the inductor $v_L(t)$.



Question 3 [CO1, CO3] [Bonus 5 points]

Design a circuit that produces the transfer function:

$$H(s) = \frac{5s}{s+2}$$

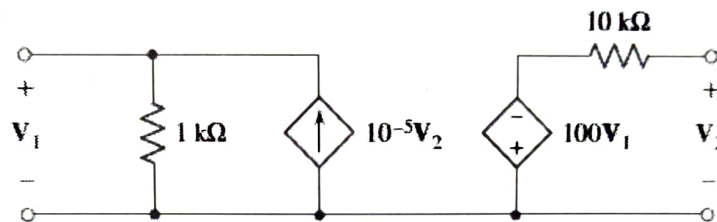
Questions 5 [CO1, CO3] [10 points]

Draw the straight-line, approximate Bode magnitude and phase plots for the following transfer function.

$$H(s) = \frac{100000s}{(s+3)(10000+20s+s^2)}$$

Questions 6 [CO1, CO3] [5 points + Bonus 5 points]

(a) Obtain the Z-parameters of the two-port network shown in the figure. [5 points]



(b) For a general two-port network, assume that the output port (right port) is connected to the voltage source V_s with series resistance of R_s . Derive the expression for input impedance Z_{in} in terms of two-port Z-parameters. (Hint: Z_{in} is the impedance seen from the input port of the network). [Bonus 4 points]

(c) Using the answers of parts (a) and (b), calculate the Z_{in} of the above drawn network if the output port is connected to a voltage source with a series 200Ω resistor. [Bonus 1 point]

Questions 7 [CO1, CO3] [10 points]

Using the Fourier series, calculate $v(t)$ across the capacitor in the circuit below. The current source $i_s(t)$ is given by the below waveform, and assume $v(0) = 0$.

